

Scent, Threshold, and Track Record: A Biomimetic Framework for Calibration-Robust Multi-Agent LLM Task Allocation

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Abstract

Many multi-agent systems are run by one central orchestrator that hands out work from the top. This puts two costs on a single node: it must judge every candidate for every task, and its own speed limits the whole system. We study an alternative, Chemotactic Task Allocation (CTA), a decentralised design inspired by biology and with no central assigner. Each task advertises a short description of what it needs (a semantic envelope). Each agent scores how well it matches, and volunteers only when that match clears a set threshold, the activation barrier. Among the volunteers, the winner has the highest Binding Energy: a score that rewards a good match and a strong track record and divides by cost. Before the winner may write anything, an integrity gate checks that it is reliable and in scope. This design leans on one assumption: that an agent can judge its own fit. That is exactly where language-model agents are known to be unreliable, so we do not assume the self-rating is correct; we treat it as a variable and vary it on purpose. Across three settings, a pre-registered protocol on synthetic populations, a fleet tuned to measured model calibration, and a live pilot with real coding agents, we find the same story. Miscalibration (an agent misjudging itself) is the main way self-selection fails, and a simple correction wins back most of the loss: discount each self-rating by the agent’s observable track record, which needs no privileged information. The gate then guards against agents that try to game the system. On scaling, the decentralised design keeps peak per-node load flat (fitted growth exponent 0.0), while a central scheduler grows as $N \times M$ (exponent 2.0) out to ten thousand agents, a roughly $250\times$ saving in coordinator cost at representative prices. We are careful about the quality claim: against a full-information optimum, which no real deployment could build, CTA does not clear its pre-registered margin, and we report that honestly; against the information-bounded central scheduler a deployment would actually run, decentralised self-selection matches it when information is fresh and pulls ahead as the central reliability table goes stale. Finally, on real Claude agents across three model families, the same two ideas become a cost result. A task wrapper (an explicit interface contract) lifts a weak model to a frontier model’s completion, and it is what lets modules built by different agents fit into one working project. A calibrated agent wrapper then routes each task to the cheapest capable model, holding frontier-level completion at roughly one fortieth of the always-frontier cost, but only when the tasks are wrapped, so the two wrappers are complementary. Deployed as an Agent Client Protocol broker, the same calibrated router keeps almost all of the frontier’s completion at a fraction of its cost in a head-to-head against naive-self-report, always-frontier, and single-cheapest routing, and we demonstrate it live, end to end, on real agents. Every result is deterministic and reproduced from seeds by one command, and the real-agent results are powered (about ten agents per cell for the ladder, five for the project) with bootstrap confidence intervals.

1 Introduction

Multi-agent systems increasingly coordinate large numbers of autonomous agents across many tasks at once. A common design puts one central orchestrator in charge of handing out the work from the top down. This is

easy to reason about, but it piles two costs onto a single node. The coordinator has to judge candidates for every task, and its own throughput caps the throughput of the whole system. As the number of agents and tasks grows, that central node becomes a bottleneck and a single point of failure.

Decentralised alternatives have a long history. The Contract Net Protocol [14] spreads control through an announce, bid, and award exchange, though the award for each task still passes through a manager. Market and auction-based allocation, surveyed by Dias et al. [5] and formalised by Gerkey and Mataric [10], treats allocation as an optimisation problem and shows where decentralised methods trade optimality for scalability. Stigmergic methods such as ant colony optimisation [7] coordinate indirectly, through signals left in a shared environment, with no central assigner at all. More recently, SwarmSys carries the same swarm principle to language-model agents, coordinating through pheromone-like traces without central control [12]; we differ in adding a trust gate, a calibration correction, and a coordinator-cost scaling result. Together these show that decentralised allocation works. But three gaps recur. The award step is often still centralised for each task. A trust or reliability check before execution is rarely built in. And there is seldom a principled account of tasks that simply cannot be done with the agents available.

We study a decentralised framework, Chemotactic Task Allocation (CTA), that closes those three gaps. Its selection principle comes from two places in biology. From cryptic female choice it borrows the idea that selection continues after the first encounter, biased by chemical signals rather than settled by a single authority [8]. From the response-threshold model of division of labour in social insects it borrows the activation barrier: an individual takes on a task only when the task’s stimulus passes its personal threshold [2]. In computing terms, each task carries a wrapper (the scent envelope) that reads an agent’s role, skills, and prompt against what the task needs, and turns that into a compatibility score $c \in [0, 1]$. Agents then self-select. An agent may take a task only when its compatibility reaches the task’s activation energy ($c \geq E_a$). Among the agents willing to take it, the winner is the one with the highest Binding Energy, $B = c \cdot \tilde{C}/L$: a score that rewards a good match and a strong record and divides by cost. Selection runs in two stages, a yes/no eligibility filter and then the activation barrier, followed by a separate trust stage, the Rejection Gate. We use the biology as design intuition, not as a literal specification.

It helps to compare CTA with a simple pull-based work queue, where agents just take the best task available. CTA adds three things on top: the activation barrier as a quality floor, the Rejection Gate as a trust boundary, and the eligibility filter with its notions of infeasible and stalled tasks. Our evaluation includes a pull-based baseline (Section 2.3) to isolate these additions, so a result is not merely the effect of going decentralised.

Any design in which agents choose their own tasks rests on one load-bearing assumption: that an agent can judge its own fit for a task. Recent studies of auctions and bidding among large language model agents find that their self-reported success rates are systematically off, so an allocation built from self-reports drifts away from the full-information optimum [9], and fixing that self-confidence is still an open problem [17]. So we treat self-assessment as a first-class variable and vary it deliberately, rather than assuming it is right. In our model the compatibility score is the agent’s self-report, drawn from its true fit plus a bias and some noise, while the quality it actually delivers depends on its true fit and real competence. Miscalibration therefore corrupts the allocation without touching the ground truth. Two questions follow. First, how badly does miscalibration hurt a self-selection allocation? Second, can an observable track record (the reliability R), which needs no privileged information, win the loss back? The Rejection Gate then acts as a safety backstop. This shifts the contribution away from decentralised self-selection, which is by now well covered, and onto the calibration robustness of self-selection, which is not.

Cost-aware model selection is an active area. LLM cascades and learned routers send easy queries to a cheap model and hard ones to a strong model, cutting inference cost several-fold with little quality loss [4, 6, 13]. CTA’s agent wrapper is a router in this family, but it differs on three points. First, it corrects the routing signal for miscalibration: the bid is the model’s own self-report, discounted by an observable track record, rather than a fixed capability estimate or a preference-trained classifier, so it stays honest even

when a model is wrong about itself, which Fradkin and Krishnan [9] shows is common. Second, it routes decomposed work: together with the task wrapper’s interface contract, it assembles a single project from modules built by different models, which whole-query routers do not do. Third, it runs on a decentralised substrate whose coordination cost stays flat as the fleet grows. Closest to the correction itself, recent work calibrates confidence online for selection: MARGIN learns per-agent, per-band calibration from the task stream for a coordinator deciding which agent to trust [1], and UCCI maps token-level uncertainty to a cost-optimal cascade threshold [11]. Both are centralised, a single coordinator or cascade decides; our contribution is not the online correction on its own but placing it in a decentralised, protocol-native router with an integrity gate against reputation-gaming, and reporting a live head-to-head against these alternatives (Section 4).

Contributions.

1. A decentralised allocation framework in which tasks advertise semantic envelopes and agents self-select by affinity, reducing per-task coordination to an atomic claim rather than a central assignment.
2. A two-stage selection model that separates categorical capability (eligibility) from graded affinity (activation energy), and yields a principled distinction between infeasible tasks (no eligible agent) and stalled tasks (eligible agents present, none clearing E_a).
3. A reliability and integrity trust gate that screens degraded and out-of-scope agents before they gain write access, evaluated as a safety backstop against adversarial agents.
4. An evaluation methodology comparing the framework against centralised baselines on scaling and match quality, including an information-bounded central scheduler as the fair, beatable comparison the full-information optimum cannot provide, and a token/dollar cost model that turns coordinator work into money.
5. A study of self-assessment miscalibration as the failure mode of self-selection: the compatibility bid is treated as the agent’s self-report, and a track-record correction (the reliability R), which uses no privileged information, recovers the completion that miscalibration costs, with the integrity gate as a safety backstop.
6. External validity in three steps: a mixed fleet whose calibration archetypes are parameterised from measured LLM behaviour (MarketBench); a live pilot with real Claude coding agents; and a capability ladder across three real model families (Haiku 4.5, Sonnet 5, Opus 4.8) that measures what the task wrapper and agent wrapper buy.
7. A self-improvement result that reads the loop as a harness [15]: a persistent, accumulating track record makes the allocation improve from its own experience toward the full-information optimum with no privileged information, while a memoryless baseline stays flat.
8. A deployed form of the router as an Agent Client Protocol broker [16], evaluated head-to-head against naive-self-report, always-frontier, and single-cheapest routing on real agent outcomes, and demonstrated in a live end-to-end vignette (Section 4).

Scope and non-goals. This is a simulation study with real-agent confirmation, not a live deployment. The atomic claim assumes a single logical coordination store providing compare-and-swap, so the framework reduces central work rather than removing it entirely. The security model covers reliability scoring and scope integrity, and extends to two forms of gaming (bid inflation and reputation sandbagging), addressed in Section 3; cross-agent collusion is left as future work.

2 Methodology

2.1 Research design

This is a comparative, controlled experiment. We measure two systems on the same task and agent populations: the decentralised framework and a centralised baseline. Both systems share the same scoring function (Binding Energy), so the only thing that differs is how allocation is coordinated. We vary the population size to test the scaling hypothesis, and we run each configuration over many seeded replications so we can report intervals rather than single numbers. We use two modes. A simulation mode runs many synthetic agents in Python, which gives scale, determinism, and cheap sweeps. A real-swarm pilot uses Claude Code subagents competing over a shared store (SQLite), where the atomic claim is a genuine compare-and-swap; this gives ecological validity.

2.2 Formal framework

This subsection states the model precisely. The list below defines each step in order, from checking whether an agent is even allowed to take a task, through scoring and firing, to the trust gate and the feedback that updates an agent’s record. There are a set of agents A of size N and a set of tasks T of size M . Each task carries a scent envelope with a domain, an activation barrier E_a (default 0.2), and a priority. Each agent has a capability profile, permissions, tools, and a reliability history.

Eligibility (binary). $\text{elig}(a, t) = \mathbf{1}[\text{ReqTool}_t \subseteq \text{Tool}_a] \cdot \mathbf{1}[\text{scope}_t \subseteq \text{Perm}_a]$: tools and permitted scope are hard requirements. The eligible set is $A(t) = \{a : \text{elig}(a, t) = 1\}$; the task is *infeasible* when $A(t)$ is empty.

Compatibility and reliability. Compatibility $c(a, t) \in [0, 1]$ is produced by the task wrapper from the agent’s role, skills, and prompt against the task requirements. Reliability $R(a) = (s_a + 1)/(n_a + 2)$ is a Laplace-smoothed success ratio over a sliding window of W recent attempts. Effective capability is $\tilde{C}(a, t) = C(a, t) \cdot R(a)$.

Selection score. The Binding Energy $B(a, t) = c(a, t) \cdot \tilde{C}(a, t) / \max(L(a, t), \varepsilon)$, with cost $L > 0$ and floor $\varepsilon = 0.01$, ranks the agents that have cleared activation; it does not gate firing.

Activation (firing). Activation drive $\Delta(a, t) = c(a, t) - E_{a,t}$. Firing probability $P_{\text{fire}} = 1$ when $\Delta \geq 0$ and $P_{\text{fire}} = \exp(\Delta/T)$ when $\Delta < 0$, with temperature $T \geq 0$; the deterministic threshold is the $T \rightarrow 0$ limit. The task is *stalled* when $A(t)$ is non-empty and the firing set is empty.

Claim and trust. The winner is $a^*(t) = \arg \max_{a \in F(t)} B(a, t)$, committed by an atomic compare-and-swap. The Rejection Gate admits a^* when $R(a^*) \geq \tau$ (default 0.6) and the action is in scope; otherwise it deflects.

Outcome and feedback. Realised quality $Q(a, t) = \text{clip}(g(S_{\text{true}}, C_{\text{true}}) + \xi)$ is a ground truth independent of the agent’s self-estimate; the attempt succeeds when $Q \geq q_{\min}$, updating (s_a, n_a) and hence R . Self-assessment acts on noisy estimates $\hat{c} = \text{clip}(c + \eta)$ with $\eta \sim \mathcal{N}(b, \sigma^2)$ for bias b and noise σ ; activation and selection use the estimates, outcomes use the truth. Annealing relaxes $E_{a,t} \leftarrow \max(E_{a,\min}, E_{a,t} - \delta)$ per waiting round.

We keep two costs separate. Total evaluation work is $W_{\text{eval}} = \sum_t |A(t)|$ (worst case $O(NM)$): the work of judging candidates. Coordinator work is $W_{\text{coord}} = \sum_t |F(t)|$: the claim attempts that actually contend at the shared store. It is the second cost, not the first, that a central node concentrates.

2.3 Baselines

The centralised baseline is a single scheduler that assigns tasks using the same scores. We use three baselines. `central_best` is the full-information quality optimum for CTA’s non-exclusive setting: it gives each task the agent with the highest expected realised quality (true fit $\times$ true capability), with agent reuse allowed. It is the fair upper bound CTA approaches from below, and no real deployment could build it because it needs the ground truth (reference for H2). `central_bounded` is the honest scheduler a deployment would actually run: it allocates from self-reported compatibility and a reliability estimate that lags, because a central table is synchronised in batches; a staleness parameter blends the current reliability with an out-of-date value (reference for H9). The third baseline is decentralised: a pull-based work queue with no barrier and no gate, which isolates CTA’s specific mechanisms from the effect of decentralisation alone.

2.4 Analysis

We report means with 95% confidence intervals. For the quality and completion metrics, which are bounded and skewed, the interval is a bootstrap percentile interval rather than a normal approximation. Each headline comparison reports how many seeds per arm a power calculation ($\alpha = 0.05$, power 0.8) requires. For scaling, we fit the growth of coordinator work against N in log-log space and report the exponent with a bootstrap CI. We prefer non-parametric tests (Mann-Whitney U) and report effect sizes alongside p -values. We fix the hypotheses and acceptance margins in advance, and we correct the secondary p -values for multiple comparisons with a Holm-Bonferroni procedure.

2.5 Validity and threats

The main threat is external validity: a simulation leaves out the variable latency, cost, and unpredictable output quality of live agents, and the miscalibration of an agent scoring itself. We mitigate this at two levels: a fleet whose calibration is set from the stated-versus-realised gaps MarketBench measures, and a live pilot with real Claude agents (Section 3). Reported failure rates for real multi-agent systems are high, and they are driven by specification ambiguity and coordination breakdown that a simulation does not reproduce [3]. For generalisability, we re-run the pre-registered hypotheses under a second, structurally different generative family. For reproducibility, the code, seeds, and configurations are versioned, and the deterministic path reproduces exactly.

3 Results

One command generates these results over the full protocol (20 seeds, agent counts from 50 to 10,000), and we release the raw per-seed rows behind every aggregate as a flat CSV. The results tell one story. First, the decentralised design removes the coordination bottleneck (H1). Against the scheduler a deployment would actually run, it matches that scheduler and then overtakes it as the scheduler’s information goes stale (H9). It pays for this with a deliberate, bounded quality cost against a full-information optimum it does not claim to beat (H2, H6). The load-bearing risk sits in self-selection itself: it rests on each agent’s own self-report, and self-reports are miscalibrated. We show that this miscalibration is the failure mode (H7), that a track-record correction using no privileged information wins back what it costs (H8), and that an integrity gate is the safety backstop (H4). We then confirm the mechanism on real Claude agents and turn it into the product claim. The task wrapper raises a weak model to frontier completion (H11), and the calibrated agent wrapper delivers frontier completion at a fraction of the cost (H12). Run over many rounds, the same loop improves from its own experience (H13).

Table 1: The whole study at a glance.

Layer	Claim	Headline result
Scaling	H1	Peak per-node load flat (exponent 0.0) vs quadratic central (2.0) to 10,000 agents
Quality	H2, H6	Reaches $\approx 94\%$ of the fair optimum; the shortfall is mostly the deliberate latency trade
Fair coordination	H9	Level with a fresh central scheduler, decisively ahead (0.88 vs 0.64) as its table goes stale
Calibration failure	H7	Raw self-report winners over-predict success by ≈ 0.20 (Brier/ECE ≈ 0.25)
Calibration fix	H8	Track-record correction recovers completion $0.37 \rightarrow 0.76$ ($p < 0.001$), no privileged information
Safety	H4	Integrity gate cuts out-of-scope violations by $\approx 90\%$ under adversarial agents
Routing/liveness	H3,H5,H10	Barrier routes every subtask to a correct specialist (1.00 vs 0.25 floor); annealing bounds stall
Task wrapper	H11	Interface contract lifts the weak model $0.950 \rightarrow 0.986$ completion (ten agents/cell, tight CIs)
Agent wrapper	H12	Calibrated routing holds frontier completion at $\approx 1/40$ of the always-frontier cost, when wrapped
Self-improvement	H13	With a persistent record, completion climbs $0.36 \rightarrow 0.84$ over rounds toward the oracle; raw stays flat

3.1 Hypothesis verdicts

Table 2 records the verdict for each hypothesis. Of eleven pre-registered synthetic hypotheses, nine are supported and two (the quality claims against a full-information optimum, H2 and H6) are reported as honest negatives.

3.2 Adversary robustness (supporting analyses)

The track record demotes an agent that games its bid. Consider a strategic adversary that inflates its self-report while it is genuinely incompetent. Starting from a clean record, it is demoted as its real failures pile up, and its win share falls from ≈ 0.22 to near zero over the rounds. A harder threat is a *sandbagging* adversary that games the reputation itself: it does honest work to earn a high reliability, then defects. Under a plain cumulative record this exploits a real lag. A recency-weighted reliability window ages the honest record out and collapses the adversary’s share the round after it defects, cutting the damage by $\approx 40\%$ (Figure 3). No reactive record can pre-empt the first defection; it can only bound it. An exposure cap limits how many tasks a single newly-defecting agent can sweep in the round it turns, from ≈ 17 failed wins with no cap to ≈ 4 at a cap of one (Figure 4). The scope-based integrity gate is the complementary backstop that does not depend on reputation at all.

Generalisation of the wrapper lift. To test whether the task-wrapper lift generalises beyond the primary tier, we built a second, structurally independent tier of eight different trap-dense tasks, disjoint from the primary tier, and ran Haiku and Sonnet under the same bare-versus-wrapped design at three agents per cell. Both models solved every task in both conditions (completion 1.00, lift 0.00). This is an honest bound rather than a null: the wrapper repairs specific, real failures (on the primary tier the $+0.036$ Haiku lift came from one genuine failure mode), and where the model is already reliable the contract is a no-op with no regression, a safety net rather than a generic booster.

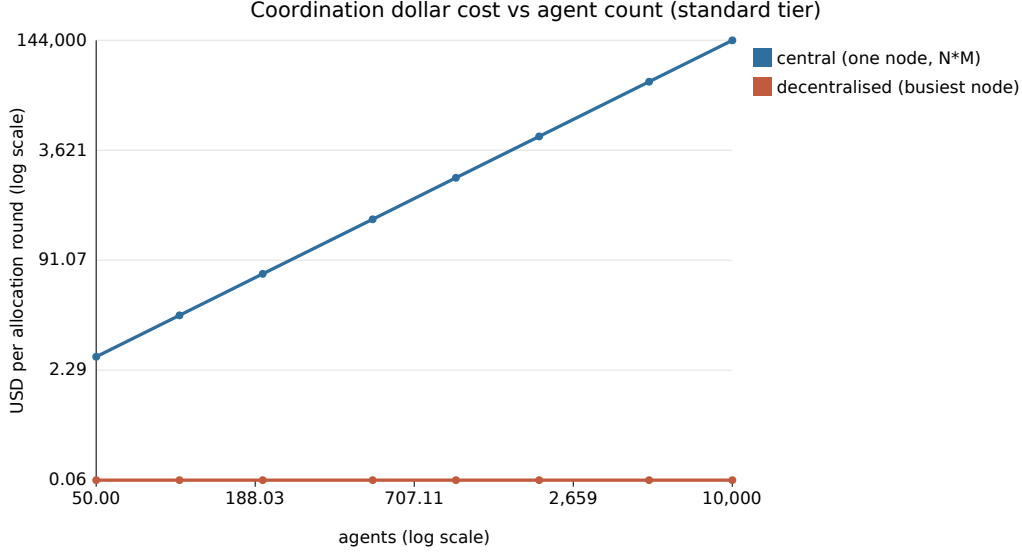


Figure 1: Scaling (H1). The central scheduler’s coordination bill grows quadratically in fleet size while the busiest decentralised node stays flat, so the two diverge by orders of magnitude out to ten thousand agents. Prices are representative tiers.

Generalisability across generators. We re-ran the population-dependent hypotheses under a structurally different latent family (smooth cosine compatibility, no skill gate). The verdicts hold under both families. The calibration recovery is also robust to how the miscalibration is generated: on a population whose per-agent bias is drawn from the measured MarketBench archetype mixture, the correction still recovers completion by ≈ 0.48 . The full verdicts under both the domains family and the latent family are summarised in Table 3.

3.3 Real agents: the two wrappers

To observe miscalibration rather than just model it, we had real Claude coding agents state their confidence on coding micro-tasks with hidden edge-case tests, then solve them one-shot with no tools. Across a nineteen-task suite over two model families (114 attempts), every attempt passed, at a mean stated confidence of ≈ 0.93 . So both families were underconfident (overconfidence gap ≈ -0.07), matching the direction MarketBench reports for the Claude models. The sign of the bias depends on the model. That is exactly why the correction reweights by an observable track record: it does not need to know the direction in advance.

To separate model capability from what the wrappers buy, we built a harder expert tier of eight trap-dense tasks and ran three model families (Haiku 4.5, Sonnet 5, Opus 4.8) under two prompts: a bare prompt and a task-wrapped prompt. Over ten Haiku agents per cell, the task wrapper lifts Haiku from 0.950 (CI [0.90, 0.988]) to 0.986 (CI [0.958, 1.00]), and leaves the already-saturated stronger models unchanged (Figure 5). The agent wrapper is Binding-Energy routing across this ladder. Over task-wrapped tasks it holds completion at 0.986 for $\approx 1/47$ of the always-frontier cost; over bare tasks it reaches 0.950 for $\approx 1/50$ (Figure 6). The two wrappers are complementary: the task wrapper raises the weak model’s fidelity, and the agent wrapper turns that into a cost saving. The ladder rests on ten agents per cell with bootstrap CIs; the project on five.

To test the wrappers on software that has to be decomposed, rather than on flat tasks, we built *miniquery*: a five-module in-memory query toolkit with a dependency graph and a hidden contract. Each model family built the whole project under the bare and the task-wrapped spec. Under the bare spec every model failed. Each one made a self-consistent but different interface choice, so the modules did not fit together, and

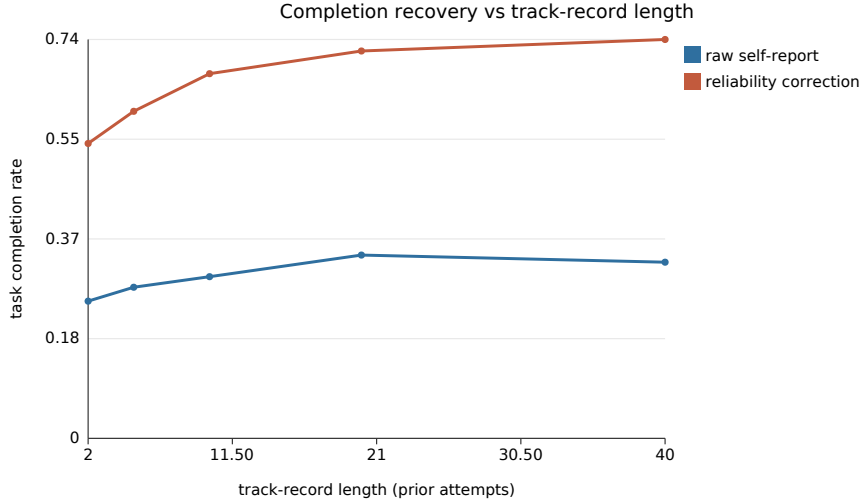


Figure 2: The calibration fix (H8). Discounting each self-report by the agent’s observable track record recovers the completion that miscalibration costs, and the recovery needs little history: even a short record recovers most of the gap.

completion was zero for all three (Figure 7). A calibration signal falls out of the ambiguity: the stronger models correctly lowered their confidence (Sonnet 0.43, Opus 0.62), while the weakest stayed overconfident (Haiku 0.89, on a two-of-five result). The task wrapper is the exact interface contract, and with it every model delivered a fully conforming project (completion 1.0). The agent wrapper then assembles the cross-model project for $\approx 1/39$ of the always-frontier cost, whereas the bare assembly fails at any price, because no model produced a conforming render without the contract.

4 Deployment: a calibration-robust routing harness

The agent wrapper is not only an experimental knob; it deploys as a live router. We present it as an Agent Client Protocol (ACP) agent [16], a JSON-RPC protocol that lets any editor talk to any coding agent. On each turn the broker elicits a confidence bid per candidate model, corrects it by the track record, gates it, routes to the cheapest model that clears the barrier, forwards that model’s reply to the editor, and records the outcome. Read as a harness in the sense of Weng [15], the software around the base models that decides how they are chosen and what is remembered, the persistent track record is what lets the deployed router improve across sessions rather than start cold each time (the deployment form of H13). Recent work calibrates confidence online for a centralised coordinator or cascade [1, 11]; our router differs in being decentralised, protocol-native, and gated against reputation-gaming.

4.1 Head-to-head against the alternatives

Routing is a decision over outcomes, so we evaluate it on the ladder outcomes already collected, at no extra cost, by leave-one-replicate-out cross-validation: reliability and self-reports are estimated from the training replicates, and each policy is scored on the held-out replicate it did not see. We compare four policies: CTA-corrected routing, naive self-report routing (the raw bid, no correction), always-frontier, and single-cheapest. Table 4 and Figure 8 give the result on the expert bare tier over ten folds. The corrected router retains 0.988 completion, within 0.013 of always-frontier, at $\approx 25\times$ lower cost, and beats naive self-report and single-cheapest by $+0.037$. One effect shows up only with real self-reports: because Claude’s stated confidence is uniformly high, naive self-report routing collapses into single-cheapest, so the track-record

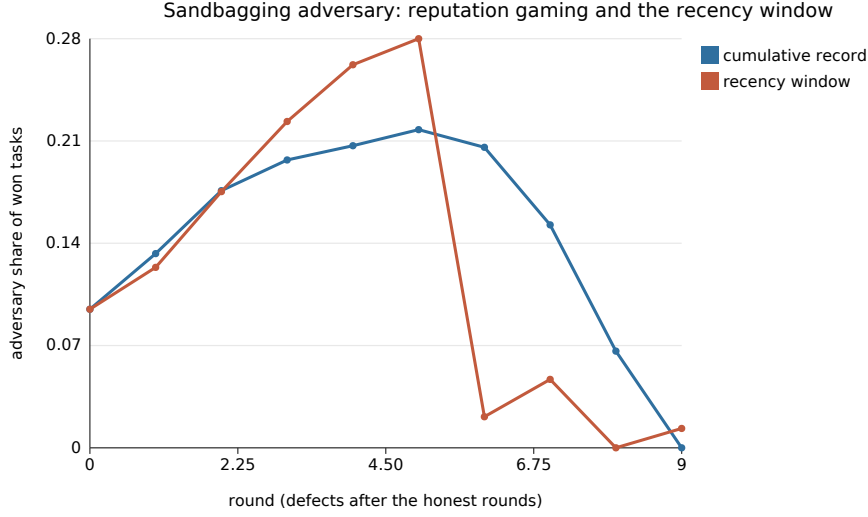


Figure 3: The sandbagging adversary builds a clean record over the honest rounds, then defects. Under a cumulative record its win share coasts down over several rounds; a recency-weighted window collapses it the round after defection.

correction is exactly what restores the ability to tell hard turns from easy ones.

4.2 A live end-to-end vignette

To show the broker running rather than replayed, we ran it live over four real Claude subagent turns, with the track record warm-started from the ladder (Table 5, Figure 9). Three tasks routed to the cheap model (Haiku) and passed. The fourth, `is_match`, is the one task Haiku is unreliable on (reliability 0.70), so its corrected bid falls to 0.595, below the 0.70 barrier, and the broker escalated to Sonnet, which passed. Routed spend was \$0.045 against \$0.72 for always-frontier, and every turn passed: the broker protected completion where the cheap model was weak and paid the cheap price everywhere else.

5 Discussion

The runs support the scaling and expressiveness claims, and the repositioning around calibration turns the earlier soft spots into results. The quality claim H2 does not hold against a fair optimum, and we report that as it stands. On scaling: once we measure the bottleneck as peak per-node load and let each agent observe only a bounded sample of tasks, CTA’s peak load is flat in the population size, while the central scheduler grows with $N \times M$. A log-log fit recovers the two growth orders cleanly. On quality: against the fair full-information optimum (`central_best`, agent reuse allowed), CTA reaches $\approx 94\%$ of the optimum and is level with pull-based. Breaking the gap down, about three quarters of it is quality the cost-aware bid trades for lower latency by design, and only about a quarter is the residual from ranking on a noisy track record. So CTA is not failing to find quality; it is optimising quality per unit cost.

The central result is about the assumption self-selection rests on. When the compatibility bid is the agent’s own self-report and competence varies, the raw self-report auction is competence-blind: completion falls to ≈ 0.37 , while the winners over-predict their own quality by ≈ 0.20 (H7). The fix is cheap and needs no privileged information. Discounting each self-report by the observable track record raises completion to ≈ 0.76 , a recovery of ≈ 0.39 that stays highly significant after multiple-comparison correction (H8). We also argue the H6 comparison against a full-information optimum is the wrong test for a deployment, because that optimum cannot exist in production. H9 replaces it with the bounded central scheduler a deployment would

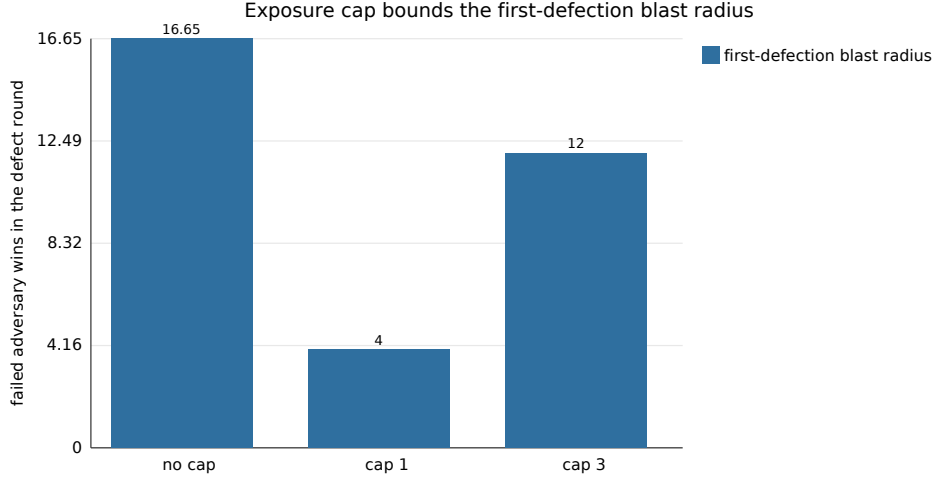


Figure 4: Failed adversary wins in the defection round fall from ≈ 17 with no cap to ≈ 4 at a cap of one task per agent per round. The cap bounds how much a single newly-defecting agent can sweep before the recency window and the gate react, at a coverage cost.

actually run. CTA is level with that scheduler when its table is fresh, and pulls decisively ahead as the table goes stale, because a decentralised agent never pays the synchronisation lag a central table does.

On external validity, the miscalibration is not just a synthetic knob. A fleet parameterised from Market-Bench reproduces the effect, and a live pilot with real Claude agents confirms on real data that a coding agent’s self-reports are miscalibrated. On solvable in-distribution tasks the direction is underconfidence, so a naive auction would under-select capable agents rather than over-select bad ones. The harder tier did not flip that sign, so the overconfident failure region is out-of-distribution here; reaching it would need tasks beyond a model’s competence, or a model family that is overconfident in-distribution. This loop is a *harness* in the sense recent work gives the term [15]: the software around a base model that decides how it plans, what it remembers, and how its work is judged. That work’s diagnosis is that a self-improving loop is only as good as the signal it optimises, and the evaluator is the weak link. CTA’s track-record correction turns an unreliable self-report into a trustworthy allocation signal, and H13 makes the self-improvement concrete. We therefore claim CTA as the calibration-robust signal layer of such a harness, not as a full self-improving system.

The scaling result has a direct commercial reading. Turn the pair-evaluation count into money at representative prices. A central scheduler that scores every agent-task pair pays a bill that grows as $N \times M$. A decentralised fleet caps each agent at a bounded sample, so the busiest node’s bill is flat and the total grows only linearly. At the standard tier the central cost per allocation round rises from $\approx \$14$ at a hundred agents to $\approx \$144,000$ at ten thousand, whereas the decentralised total is $\approx \$576$ and no single node pays more than $\approx \$0.06$: a $\approx 250\times$ saving at the top of the range. That shape, a quadratic central bill against a flat per-node decentralised one, is a property of the coordination structure, not of the prices.

Practical implications. The immediate use is cost control for multi-model coding assistants and agent platforms, the same setting cost-aware routers already target commercially [4, 6, 13]. What the correction adds there is specific and asymmetric: a cheap model that is confidently wrong is expensive, because a plausible but broken answer costs review, rework, and lost trust far more than the tokens it saved, and the correction escalates exactly those turns while keeping the cheap price everywhere else. Two properties decide whether it is worth deploying. First, the gain is failure-contingent: it grows with how often, and how overconfidently, the cheap tier fails, so the setting that rewards it most is a heterogeneous fleet of small,

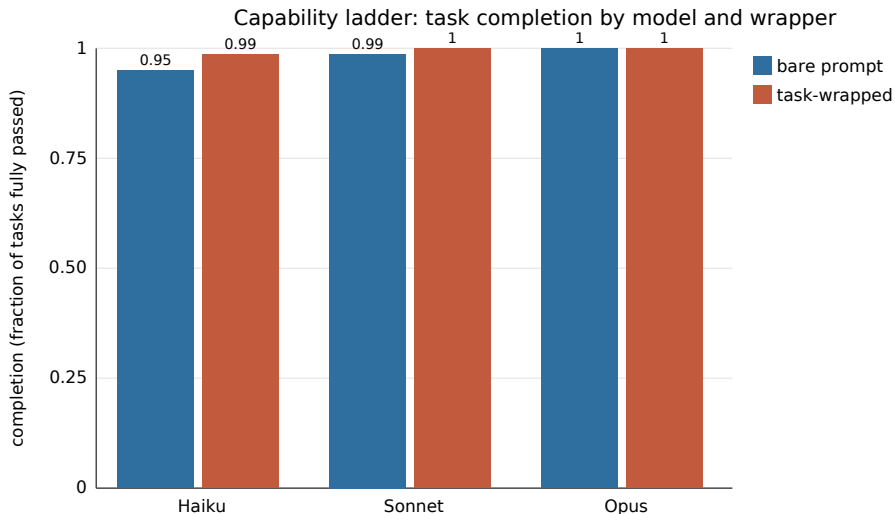


Figure 5: The task wrapper (H11) lifts the weakest model to the frontier’s completion and leaves the already-saturated models unchanged. Bars are completion; the wrapper helps most where the model is weakest.

cheap, often overconfident models, not a fleet of already-reliable frontier tiers (our underconfident Claude result sits near the no-regression floor, not the ceiling). Second, the defensible part is not the correction itself, which is simple and already appears in concurrent work [1, 11], but the two things around it: a persistent track record that makes the router cheaper and more reliable the longer it runs, from ordinary use and with no privileged information (the self-improvement result, H13); and a protocol-native deployment that drops into existing editors with no bespoke integration [16]. In short, the mechanism is close to a commodity; the accumulated calibration data and the deployment surface are where durable value would sit. We state this as an implication, not a demonstrated business result: this is a mechanism study with real-agent confirmation, and any production claim would need the head-to-head against a named router on a shared benchmark noted next.

Where this research stands. Eleven pre-registered hypotheses on synthetic populations are evaluated at full protocol scale, nine supported and two reported as honest negatives. Two further hypotheses (the task-wrapper lift and the calibrated-routing cost-efficiency) are added on real Claude agents and supported with bootstrap confidence intervals: the ladder rests on about ten agents per cell and the project on five. It is a mechanism study with real-agent confirmation, not a benchmarked production system, and every claim is marked against that boundary. The natural next step is a head-to-head against a named router on a shared, established benchmark.

6 Building the wrappers

The task wrapper. A task wrapper is an explicit interface contract. It has a name, a signature, a one-line description, a set of named acceptance criteria, a few worked examples, and a short self-check the agent runs before it commits. The orchestrator owns and builds it, which keeps it stable across a mixed fleet. You can build it from a machine-checkable spec, from examples, from a natural-language description via a cheap wrapping model, or from downstream integration failures added as new acceptance clauses. The design rule is simple: the acceptance criteria should name the properties a hidden test would check. The lift comes from

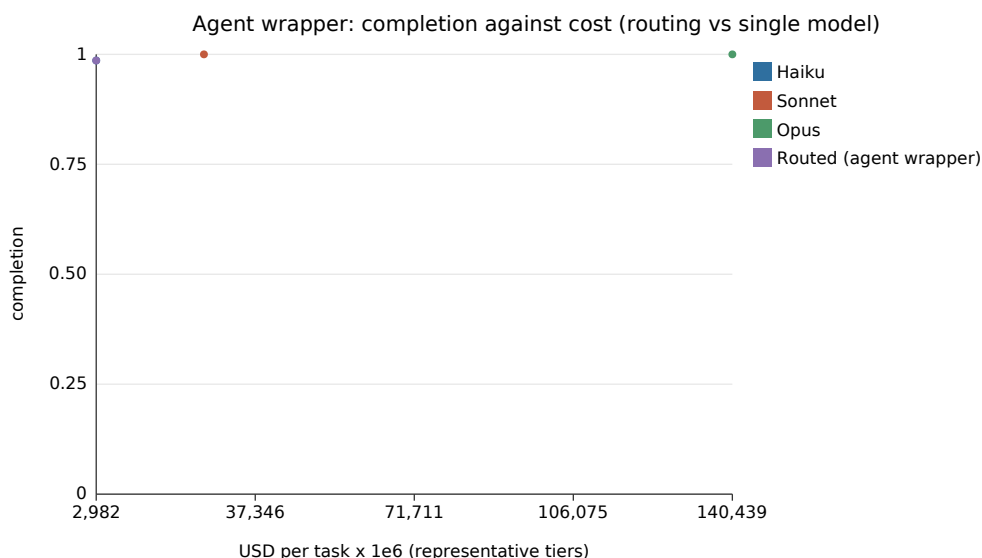


Figure 6: The agent wrapper (H12). Routing over task-wrapped tasks holds frontier completion at a fraction of the always-frontier cost; over bare tasks it leaves the cheap model’s residual gap. Cost is on a representative-tier scale.

surfacing the unstated edge cases that separate a competent-looking answer from a correct one.

Why a contract, and not just a smarter model. The wrapper does the job biology solves with a specific signal rather than a stronger worker. An enzyme acts only on a substrate whose shape matches its active site, and a hormone binds only its own receptor: the specificity comes from a precise signal, not from raw force. The acceptance criteria are that active site, the exact features a match must have. The composability result has a sharper analogy in the genetic code. A gene is a precise specification, and any ribosome, a different worker each time, reads the same sequence and builds the same protein, because they all share one exact code; loosen the specification and separately built parts would no longer fit together. Our project experiment is the same story in software: separately built modules assemble only because they share the wrapper’s contract, and a more capable model is a faster ribosome that still cannot rescue an ambiguous specification. The lesson is that coordination failure here is a specification problem, not an intelligence problem. As elsewhere in this work, we take the biology as design intuition, not as a claim about the code.

The agent wrapper. The agent wrapper turns a fleet of models into one calibrated router. Each model has a price tier; the fleet holds the models cheapest first, plus a per-task reliability table. Routing is one rule. Each model reports a self-assessed confidence. The bid is discounted by that model’s reliability on this kind of task. The task then goes to the cheapest model whose corrected bid clears the barrier, or, if none clears it, escalates to the highest corrected bid. Reliability is updated after each outcome by a moving average, so the router learns the fleet from the first real results instead of assuming a fixed capability. That is the difference from a static-capability router.

The loop. The two wrappers are not a single call but a loop, and the loop is where the results compound. The track record sharpens, so more tasks safely drop to cheaper models. The task contracts sharpen from integration failures, so weak models grow more reliable on exactly the tasks that used to break them. Calibration robustness and cost efficiency are properties of this loop, not of any one allocation.

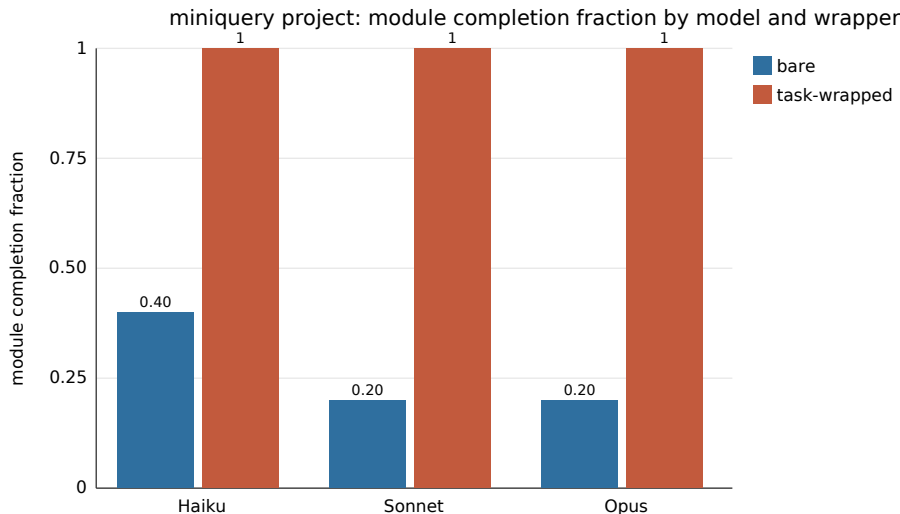


Figure 7: The dependency-graph project. Under a loose spec every model, including the frontier one, ships a divergent interface and no working project; the interface contract makes all three deliver a conforming five-module project. Capability does not rescue integration; the contract does.

7 Outlook and conclusion

The dominant pattern today is a single large model called once per task, sometimes fronted by a router that picks a model by static capability. That pattern pays frontier-token prices everywhere, and, as the project experiment shows, it cannot split a task across independent workers without an interface contract. The alternative this work supports is a swarm of microagents: small, cheap, specialised workers, each inside a tight harness, coordinated by calibrated self-selection. The claim to a breakthrough rests on combining three things usually pursued separately. First, a decentralised substrate that removes the coordinator bottleneck. Second, a routing rule that corrects the self-report by an observable track record and screens the winner for safety. Third, a task wrapper: the contract that makes a swarm’s output fit together. If the real-agent results hold at scale, we can grow agent systems by the count of cheap microagents rather than the size of one model, a different and far cheaper axis of growth.

In summary, CTA is a decentralised, biomimetic framework in which tasks advertise contracts and agents self-select by a calibrated, reliability-weighted, gated bid. It holds the busiest node’s coordination load flat as the fleet grows to ten thousand agents. It names and fixes the failure mode of self-selection, miscalibration, recovering the completion it costs with a track-record correction that uses no privileged information. And on real Claude agents, the two wrappers turn the mechanism into a cost result, holding frontier-level completion at roughly one fortieth of the always-frontier cost. The practical thesis is that a swarm of cheap, wrapped microagents coordinated by calibrated self-selection can match a single expensive model at a fraction of its cost, and that is the direction this work opens.

References

- [1] Joss Armstrong. MARGIN: Runtime confidence calibration for multi-agent foundation model coordination. *arXiv preprint arXiv:2605.22949*, 2026.
- [2] Eric Bonabeau, Guy Theraulaz, and Jean-Louis Deneubourg. Quantitative study of the fixed threshold

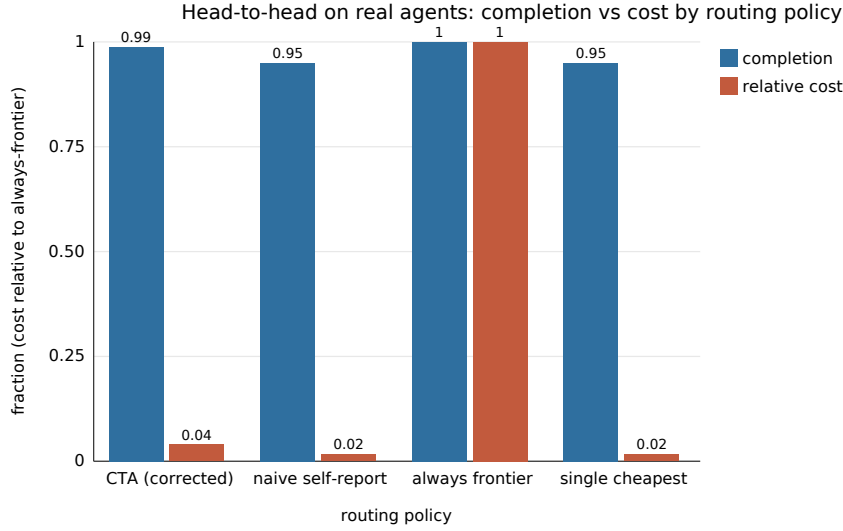


Figure 8: Completion against cost by routing policy, cost as a fraction of the always-frontier cost. The corrected router keeps almost all of the frontier’s completion at a small fraction of its cost, and separates from naive self-report and single-cheapest, which sit together.

model for the regulation of division of labour in insect societies. *Proceedings of the Royal Society of London B: Biological Sciences*, 263(1376):1565–1569, 1996. doi: 10.1098/rspb.1996.0229.

- [3] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? *arXiv preprint arXiv:2503.13657*, 2025.
- [4] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: how to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.
- [5] M. Bernardine Dias, Robert Zlot, Nidhi Kalra, and Anthony Stentz. Market-based multirobot coordination: a survey and analysis. *Proceedings of the IEEE*, 94(7):1257–1270, 2006. doi: 10.1109/JPROC.2006.876939.
- [6] Dujian Ding, Ankur Mallick, Shaokun Zhang, Chi Wang, Daniel Madrigal, Mirian Del Carmen Hipolito Garcia, Menglin Xia, Laks V. S. Lakshmanan, Qingyun Wu, and Victor Rühle. BEST-Route: Adaptive LLM routing with test-time optimal compute. *arXiv preprint arXiv:2506.22716*, 2025.
- [7] Marco Dorigo, Vittorio Maniezzo, and Alberto Colorni. Ant system: optimization by a colony of cooperating agents. *IEEE Transactions on Systems, Man, and Cybernetics, Part B*, 26(1):29–41, 1996. doi: 10.1109/3477.484436.
- [8] John L. Fitzpatrick, Charlotte Willis, Alessandro Devigili, Amy Young, Michael Carroll, Helen R. Hunter, and Daniel R. Brison. Chemical signals from eggs facilitate cryptic female choice in humans. *Proceedings of the Royal Society B: Biological Sciences*, 287(1928):20200805, 2020. doi: 10.1098/rspb.2020.0805.
- [9] Andrey Fradkin and Rohit Krishnan. MarketBench: evaluating AI agents as market participants. *arXiv preprint arXiv:2604.23897*, 2026.

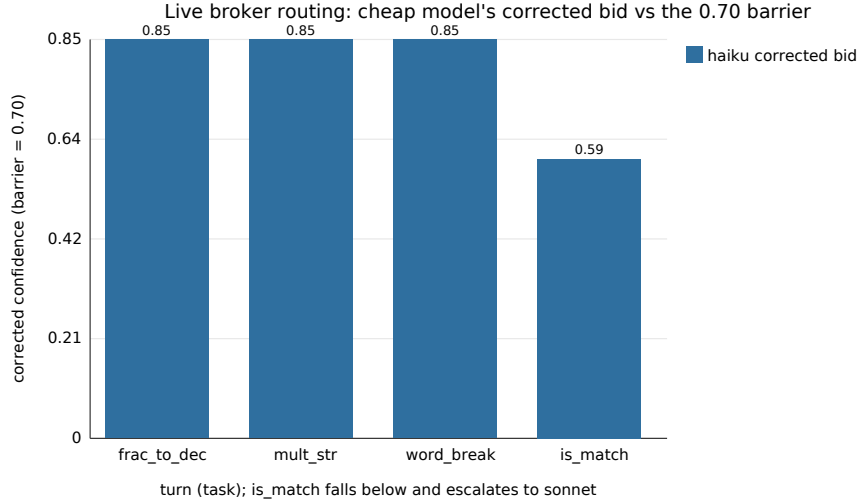


Figure 9: The cheap model’s (Haiku) reliability-corrected bid across the four live turns. It clears the 0.70 barrier on the first three tasks, which route to Haiku, and dips to 0.595 on `is_match`, which escalates to Sonnet. The dip below the barrier is the escalation.

- [10] Brian P. Gerkey and Maja J. Matarić. A formal analysis and taxonomy of task allocation in multi-robot systems. *The International Journal of Robotics Research*, 23(9):939–954, 2004. doi: 10.1177/0278364904045564.
- [11] Varun Kotte. UCCI: Calibrated uncertainty for cost-optimal LLM cascade routing. *arXiv preprint arXiv:2605.18796*, 2026.
- [12] Ruohao Li, Hongjun Liu, Leyi Zhao, Zisu Li, Jiawei Li, Jiajun Jiang, Linning Xu, Chen Zhao, Mingming Fan, and Chen Liang. SwarmSys: Decentralized swarm-inspired agents for scalable and adaptive reasoning. *arXiv preprint arXiv:2510.10047*, 2025.
- [13] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024.
- [14] Reid G. Smith. The contract net protocol: high-level communication and control in a distributed problem solver. *IEEE Transactions on Computers*, C-29(12):1104–1113, 1980. doi: 10.1109/TC.1980.1675516.
- [15] Lilian Weng. Harness engineering for self-improvement. Lil’Log, 2026. <https://lilianweng.github.io/posts/2026-07-04-harness/>.
- [16] Zed Industries. Agent client protocol (acp). <https://agentclientprotocol.com>, 2025. Open protocol (JSON-RPC 2.0 over stdio) connecting code editors and coding agents; released August 2025, Apache-2.0.
- [17] Chao Zhang, Zheng Zhu, Yi Wei, Bo Tian, Jian Liu, Hao Wang, Xin Wang, and Yang Liu. Confidence-calibrated small-large language model collaboration for cost-efficient reasoning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*, 2026. arXiv:2603.03752.

Table 2: Hypothesis verdicts. “Supported” verdicts carry the observation summarised.

Hypothesis	Verdict	Observation
H1 scaling	supported	CTA peak load flat at 32 as N grows $50 \rightarrow 10,000$; central grows as $N \times M$. Log-log exponents 0.0 (CI $[0, 0]$) vs 2.0 (CI $[2, 2]$).
H2 quality	not supported	CTA reaches $\approx 94\%$ of the fair optimum (0.883 vs 0.937), level with pull-based; does not clear the 5% margin. About three quarters of the gap is quality traded for latency by design.
H3 expressiveness	supported	Infeasible and stalled tasks labelled with full recall against ground truth.
H4 safety gate	supported	With $\approx 30\%$ adversarial agents and gate recall 0.9, violations cut by $\approx 90\%$ (1.2 vs 12.6 per run). A measured reduction, not a tautological zero.
H5 annealing	supported	Without annealing, feasible tasks never claimed; with it, every feasible task resolves at bounded stall (1–3 rounds), falling smoothly with the annealing rate.
H6 heterogeneity	not supported	Against the fair optimum CTA sits slightly below at every heterogeneity; the gap does not close as the population specialises.
H7 miscalibration	supported	Raw self-report winners over-predict realised quality by ≈ 0.20 (Brier/ECE ≈ 0.25). Structural fit-vs-competence gap, not an artefact of injected bias.
H8 track record recovers	supported	Discounting by R raises completion $0.37 \rightarrow 0.76$ under worst overconfidence (recovery 0.39, Holm $p < 0.001$ at 20 seeds).
H9 matches bounded central	supported	Level when the central table is fresh (0.88 vs 0.91); decisively ahead at full staleness (0.88 vs 0.64; advantage 0.25, Holm $p \approx 0.03$).
H10 specialist routing	supported	Barrier holds routing accuracy 1.00 at every observability (chance floor 0.25); without it, routing falls to 0.47 under tight observability.
H11 task-wrapper lift	supported (real)	Over ten Haiku agents/cell, the wrapper lifts Haiku 0.950 (CI $[0.90, 0.988]$) $\rightarrow$ 0.986 (CI $[0.958, 1.00]$); Sonnet/Opus already saturated. On the project every model fails bare (completion 0) and delivers wrapped (completion 1.00).
H12 agent-wrapper cost	supported (real)	Routing over wrapped tasks holds completion 0.986 at $\approx 1/47$ of always-Opus cost; over bare tasks 0.950 at $\approx 1/50$. Project wrapped assembly at $\approx 1/39$; bare fails at any price.
H13 self-improvement	supported	Over twelve rounds, completion under reliability selection climbs $0.358 \rightarrow 0.844$, reaching the oracle (0.767), while raw stays flat at ≈ 0.35 . The gain is the accumulating memory, not the task stream.

Table 3: Verdicts under two structurally different generators.

Hypothesis	Domains family	Latent family
H2 quality within margin	not supported	not supported
H4 safety gate reduces violations	supported	supported
H7 self-reports over-predict	supported	supported
H8 track record recovers	supported	supported

Table 4: Head-to-head on real agents (leave-one-replicate-out replay, expert bare tier, ten folds).

Policy	Completion (95% CI)	Cost per task-set (USD)
CTA (corrected)	0.988 [0.963, 1.000]	0.057
Naive self-report	0.950 [0.913, 0.988]	0.024
Always frontier	1.000 [1.000, 1.000]	1.440
Single cheapest	0.950 [0.913, 0.988]	0.024

Table 5: Live ACP-broker vignette over four real Claude subagent turns. Corrected bids are Haiku/Sonnet/Opus; the cheapest model clearing the 0.70 barrier wins.

Turn	Task	Routed to	Corrected bids (h/s/o)	Outcome
1	<code>fraction_to_decimal</code>	Haiku	0.85/0.92/0.96	pass
2	<code>multiply_strings</code>	Haiku	0.85/0.92/0.96	pass
3	<code>word_break</code>	Haiku	0.85/0.92/0.96	pass
4	<code>is_match</code>	Sonnet	0.595/0.92/0.96	pass